



PUMP FOUNDATION

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1.0 PURPOSE

This document establishes guidelines for the vibration check of small pump foundations. Design consideration for the pressure forces, potential anchor forces if any, surge/hammer system forces, operating pressures and pressure fluctuations shall be included as appropriate in the final design under static checks (e.g., soil bearing, concrete design, and reinforcement design etc.)

Scope

This document provides the following:

- A brief description of different types of pumps.
- A recommended design procedure for a soil supported foundation.
- Reinforcing requirements.
- Recommendations for grouting of pump bases.
- A chart showing coefficient of subgrade reaction K values for various types of soils (Attachment 01).
- A typical foundation sketch for a vertical (can) pump (Attachment 02).
- Design curves and examples (Attachment 03 and 04)

2.0 APPLICATION

This document should be used to design a soil supported foundation for small (skid mounted pumps smaller than 5000 lbs [2270kg] and a frequency larger than 1200 RPM) centrifugal, reciprocating and vertical pumps. Foundations for large centrifugal pumps (greater than 500 horsepower [375kW]), reciprocating pumps (greater than 200 horsepower [150kW]), more than one pump supported on one mat, inline pumps and any critical service pump shall be designed in accordance with the guidelines listed in Section 7 Resources.

3.0 INTERFACING DISCIPLINES

- Mechanical
- Piping

4.0 DEFINITIONS

Not Used

5.0 REQUIREMENTS

Not Used



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6.0 EXECUTION OF GUIDELINES

6.1 Types of Pumps

6.1.1 Centrifugal Pump

A centrifugal pump is a rotating equipment item with an impeller rotating around the shaft. The three main components are the pump casing, a motor, and a baseplate. These pumps are normally well balanced and do not have large unbalanced forces that create foundation vibration.

6.1.2 Reciprocating Pump

A reciprocating pump is a positive displacement type pump with varying numbers of pistons creating a thrust. The three main components are the pump casing, a motor, and a baseplate. These pumps have a varying magnitude of unbalanced forces; and in general, the design is treated like a small compressor. These types of pump involve a crank mechanism which produces primary and secondary forces and a couple based on cylinder size and arrangement. The foundation should be sized to avoid resonance.

6.1.3 Rotary Pump

A rotary pump is a positive displacement pump consisting of a fixed casing containing gears, cams, screws, vanes, plungers or similar elements actuated by rotation of the drive shaft.

6.1.4 Vertical (Can) Pump

A vertical (can) pump is an equipment item with a propeller revolving around the vertical axis and a housing extending below the base. These pumps are supported vertically in a pit or in a pipe casing as shown in Attachment 02. Vertical pumps are centrifugal, foundation analysis shall be determined based on the size and rating of the pump.

6.1.5 In-Line Pump

An in-line pump is a very small vertical centrifugal pump generally supported by the piping system. A small concrete base, as shown in Attachment 06 In-line Pump Foundations, is often provided to eliminate weight imposed on the piping system.

6.1.6 Spring Mounted Pump

A spring mounted pump is either a centrifugal or a reciprocating pump with a base supported on springs which in turn rest on a concrete foundation. These pumps are used for very inflexible piping systems and are very rarely used. Foundation design for this type of pump should be in accordance with the recommendations of the spring isolator manufacturer.

6.2 Design (Soil Supported Foundation)

6.2.1 General

This document outlines a recommended design procedure for a soil supported pump foundation.



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To size the pump foundation, comply with the following parameters:

- Provide minimum edge distance for anchor bolts of 6 times the bolt diameter.
- Extend foundation a minimum of 2 inches [50 mm] around the pump base frame or the oil pan, if one exists.
- Extend the required depth below the frost line where applicable or minimum 2 feet [600 mm] below the high point of finish surface elevation.
- Provide a mass ratio of 3 to 1 for centrifugal pumps and 5 to 1 for reciprocating pumps.
- For large pumps, obtain the anchor forces, surge forces or any other upset condition forces.
- Check soil bearing for pump weight plus any other static forces using a limit of 50 percent of the normal static allowable bearing pressure.

The rational method for sizing the pump foundation is to provide the minimum parameters listed above, check soil bearing and then to calculate the frequency ratio to confirm that it is outside the resonance range.

Typically the frequency ratio R shall be outside the range of 0.7 to 1.4. Because most pumps operate between the speed of 1,200 to 3,600 RPM (revolutions per minute), the frequency ratio of 1.4 or more is recommended to avoid resonance and to keep the foundation overtuned. An overtuned foundation avoids the resonance condition potentially generated during start-up and shut down of the pump.

The frequency ratio R is defined:

$$R = \frac{F_{pump}}{F_n}$$

Where

F_{pump} = Pump Operating Speed, RPM

F_n = Natural Frequency of Pump-Foundation-Soil System, CPM (cycles per minute)

The natural frequency of the pump-foundation-soil system varies with the soil subgrade modulus K that supports the foundation and the total weight at the soil surface.

Attachment 01 lists the textbook values of the soil subgrade modulus K values. However, the soil subgrade modulus should normally be provided in the project geotechnical report and verified by the Lead Engineer.

The total weight of the pump-foundation system includes the weight of the pump (including motor and baseplate) and the weight of the foundation. The weight of the pump, motor, and baseplate is given by the Supplier on the pump outline drawing. In most cases, the information given on the Supplier catalog sheets varies substantially from the information given on the certified outline. Use the latest certified outline.

The natural frequency of the pump-foundation-soil system in the vertical direction is calculated as a single mass-spring system:

$$F_n = \frac{1}{2\pi} \sqrt{\frac{k}{M}} \text{ (CPS) Cycles per Second}$$



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Substituting Soil Subgrade Modulus (K) for Stiffness (k) and Weight (W) for Mass (M) and converting from (CPS) to (CPM):

$$F_n = \frac{60}{2\pi} \sqrt{\frac{K (12)^3 A g}{W_p + W_c}} \text{ (CPM)} \quad \text{Metric : } F_n = \frac{60}{2\pi} \sqrt{\frac{K A g}{W_p + W_c}} \text{ (CPM)}$$

Where

K = Soil Subgrade Modulus, lbs/in³ [kN/m³]

A = Base Area of Foundation, ft² [m²]

g = Acceleration of Gravity, 32.2ft/sec² [9.81m/sec²]

W_p = Pump Weight, lbs [kgf]

W_c = Concrete Weight, lbs [kgf]

Substituting Gross Soil Bearing: $P_g = \frac{(W_p + W_c)}{A}$

$$F_n = \frac{60}{2\pi} \sqrt{\frac{K (12)^3 g}{P_g}} \text{ (CPM)} \quad \text{Metric : } F_n = \frac{60}{2\pi} \sqrt{\frac{K g}{P_g}} \text{ (CPM)}$$

Further reducing:

$$F_n = 2253 \sqrt{\frac{K}{P_g}} \text{ (CPM)} \quad \text{Metric : } F_n = 29.9 \sqrt{\frac{K}{P_g}} \text{ (CPM)}$$

To avoid a resonance, the pump operating speed F_{pump} should at least equal to $1.4F_n$:

$$F_{min} = (1.4)2253 \sqrt{\frac{K}{P_g}} \text{ (CPM)} \quad \text{Metric : } F_{min} = (1.4)29.9 \sqrt{\frac{K}{P_g}} \text{ (CPM)}$$

$$F_{min} = 3154 \sqrt{\frac{K}{P_g}} \text{ (CPM)} \quad \text{Metric : } F_{min} = 42 \sqrt{\frac{K}{P_g}} \text{ (CPM)}$$

Thus, for pumps with an operating speed of 1,200 RPM or greater, the simplified calculation steps for foundation design are as follows:

- Select soil subgrade modulus K , lbs/in³ [kN/m³]
- Estimate minimum concrete dimensions
- Calculate total weight ($W_p + W_c$)



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- Calculate gross soil bearing P_g , lbs/ft² [kN/m²]
- Calculate F_{min} (CPM)
- Compare pump speed F_{pump} (RPM) with F_{min}

If pump speed $F_{pump} \geq F_{min}$, the foundation is acceptable.

If pump speed $F_{pump} < F_{min}$, increase foundation dimensions and recalculate F_{min} until pump speed $F_{pump} \geq F_{min}$.

Horizontal wind and seismic loads are not usually considered in the design of pump foundations however these load conditions should be reviewed for the design and selection of the pump anchor bolts.

The above procedure only accounts for the vertical translation mode of vibration. Critical equipment foundations should be checked for the other vibration modes.

6.3 Reinforcing

Generally, pump foundations do not need reinforcing for stress requirements. Placing nominal reinforcing such as #5 at 12 inches [12 mm c/c 150 mm] each way is sufficient to prevent cracking of concrete surface at the top. The normal practice in most offices is to place bars on the top and all faces. In this case, provide #5 at 12 inches [12 mm c/c 150 mm] U-bars each way at top and enclose them with #4 U-ties [10 mm U-ties] on sides. Pump foundations placed on piles should be evaluated individually for reinforcing requirements.

As a minimum the minimum steel requirements of ACI 318 should be followed.

6.4 Grouting

In general, two types of grout are used for grouting the pump base. These are epoxy and nonshrink cementitious, or a combination of them. Sometimes the type of grout to be used is specified on the Supplier outlines. In this case, in order not to jeopardize the pump warranty, use of the specified grout is recommended. The project grout specification should be reviewed for any special client requirements.

6.4.1 Pump Base

The pump base may consist of a conventional steel plate or an inverted trough fabricated using small channels around the perimeter with a steel plate on top. The cover plate supports the pump and may have holes for grouting the space between the cover plate and the concrete foundation. In either case, proper grouting of the pump bases is essential for the long term, successful operation of the pump.

6.4.2 Epoxy Grout

Epoxy grout is a mixture of resins with an exothermic characteristic; it is nonshrink in nature. Although expensive, because of its flowable characteristic, this grout fills all voids under the base plate. The result is a reliable pump performance. If the grout thickness exceeds 4 inches [100 mm], the amount of heat generated requires taking special precautions as specified by the



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manufacturer to eliminate the possibility of cracking the base plate which may require the addition of 3/8" aggregate (i.e. pea gravel).

6.4.3 Nonshrink Cementitious Grout

Nonshrink cementitious grout is commonly used for pump bases. As the name indicates, the grout is a cement based mixture that is compatible with concrete. It has a nonshrink characteristic when cured and leaves no voids under the base due to shrinkage. This type of grout is flowable, non-exothermic, and less expensive than epoxy grout. It is commonly used for deep pump bases.

6.4.4 Combinations of Grouts

Sometimes, to economize on large projects and to minimize heat generation for deep pump bases, pump bases are grouted with a combination of layers of an epoxy grout and nonshrink cementitious grout. After the pump is aligned and ready for grouting, pour a layer of epoxy grout to fill the gap between the top of foundation concrete and the bottom of the base channel frame.

Once the epoxy has hardened, the space between the top of the epoxy and the underside of the cover plate is completely filled with nonshrink cementitious grout or filled to 1 inch [25 mm] below the underside of cover plate with sand cement grout.

The grouting procedure of the pump bases is generally dictated by the Client and is defined in the project specification for grouting. Note!!! If the Client criteria differs from the PIP Guideline for Grouts STS3600/3601, modify the project specific version of the specification.

7.0 RESOURCES

Guidelines

Guideline 000.215.1233	Vibrating Machinery Foundations on Soil (Using Frequency Independent Criteria)
Guideline 000.215.1234	Vibrating Machinery Foundations on Soil
Guideline 000.215.1235	Vibrating Machinery Foundations on Piles

8.0 REFERENCES

Global Standards

PIP STS03600	Nonshrink Cementitious Grout Specification
PIP STS03601	Epoxy Grout Specification

9.0 ATTACHMENTS, ADDENDA

Attachments

Attachment 01	Modulus of Subgrade Reaction, K
Attachment 02	Typical Detail for a Vertical (Can) Pump Foundation
Attachment 03	Pump Foundation Design Curves [Imperial and Metric Units]



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Attachment 04

Sample Design 1: Pump Foundations [Imperial Units]

Attachment 05

Sample Design 1: Pump Foundations [Metric Units]

Attachment 06

In-line Pump Foundation Design Criteria

Addenda

None



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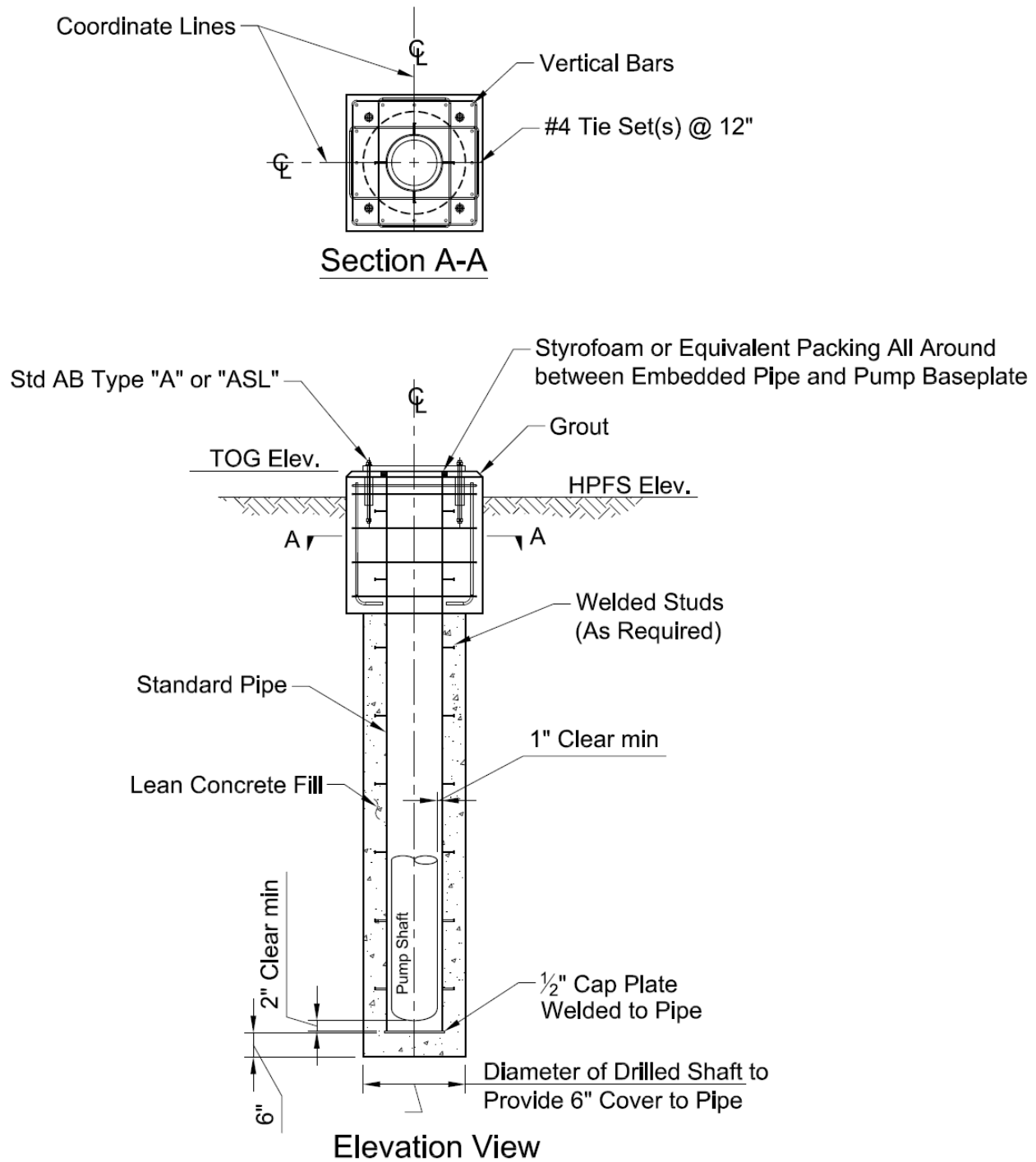
Modulus of Subgrade Reaction, K

Major Divisions	Soil Groups and Typical Description	Group Symbols	K-Value lbs/in ³	K-Value kN/m ³
Gravel and Gravelly Soils	Well-graded gravel and gravel-sand mixtures. Little or no fines.	GW	500	136000
	Well-graded gravel-sand-clay mixtures. Excellent binder.	GC	400	109000
	Poorly graded gravel and gravel-sand mixtures. Little or no fines.	GP	300	81000
	Gravel with fines, very silty gravel, clayey gravel, poorly graded gravel-sand-clay mixtures.	GF	250	68000
Sands and Sandy Soils	Well-graded sands and gravelly sands. Little or no fines.	SW	250	68000
	Well-graded sand-clay mixtures. Excellent binder.	SC	250	68000
	Poorly graded sands. Little or no fines.	SP	200	54000
	Sand with fines, very silty sands, clayey sands, poorly graded sand-clay mixtures.	SF	180	49000
Fine-grained Soils Having Low to Medium Compressibility	Silts (inorganic) and very fine sands, mo, rock, flour silty or clayey fine sands with slight plasticity.	ML	150	41000
	Clays (inorganic) of low to medium plasticity, sandy clays, silty.	CL	130	35000
	Organic silts and organic silt-clays of low plasticity.	f	100	27000
Fine-grained Soils Having High Compressibility	Micaceous or diatomaceous fine sandy and silty soils, elastic silts.	nd	60	16000
	Clays (inorganic) of high plasticity, fat clays.	fa	50	13500
	Organic clays of medium to high plasticity.	OH	40	11000



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Typical Detail for a Vertical (Can) Pump Foundation





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Pump Foundation Design Curves [Imperial and Metric]

IMPERIAL

Given:

Soil Subgrade Modulus
Pump Operating Speed
Pump Weight

Determine:

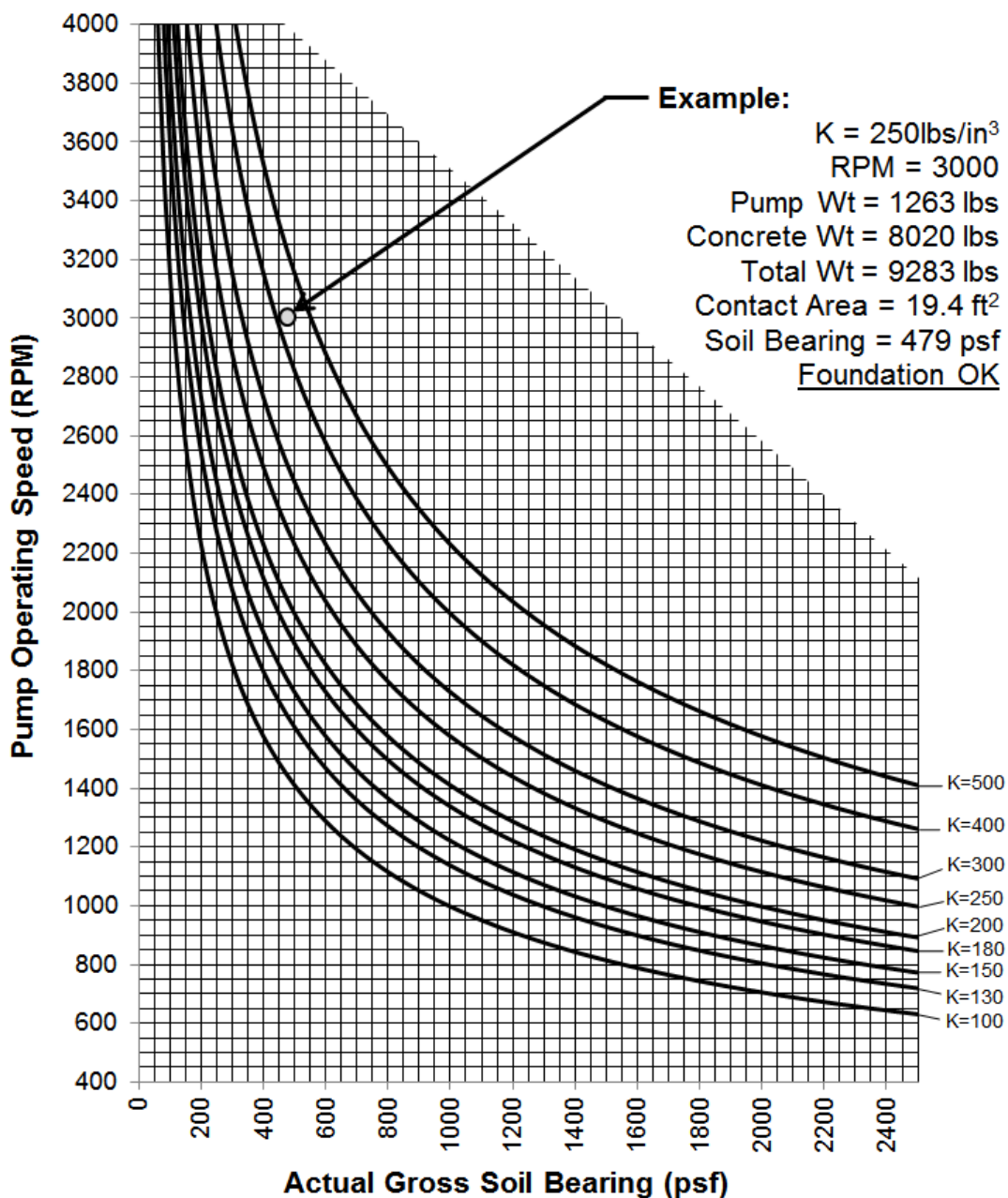
Foundation Size
Actual Gross Soil Bearing

Procedure:

Plot RPM vs Gross Soil Bearing

1. If plotted point is above the K-curve, foundation is OK
2. If plotted point lies below the k-curve, deepen foundation

If pump operates below 1200RPM or is too far below the K-curve, investigate using a spread foundation with pier.

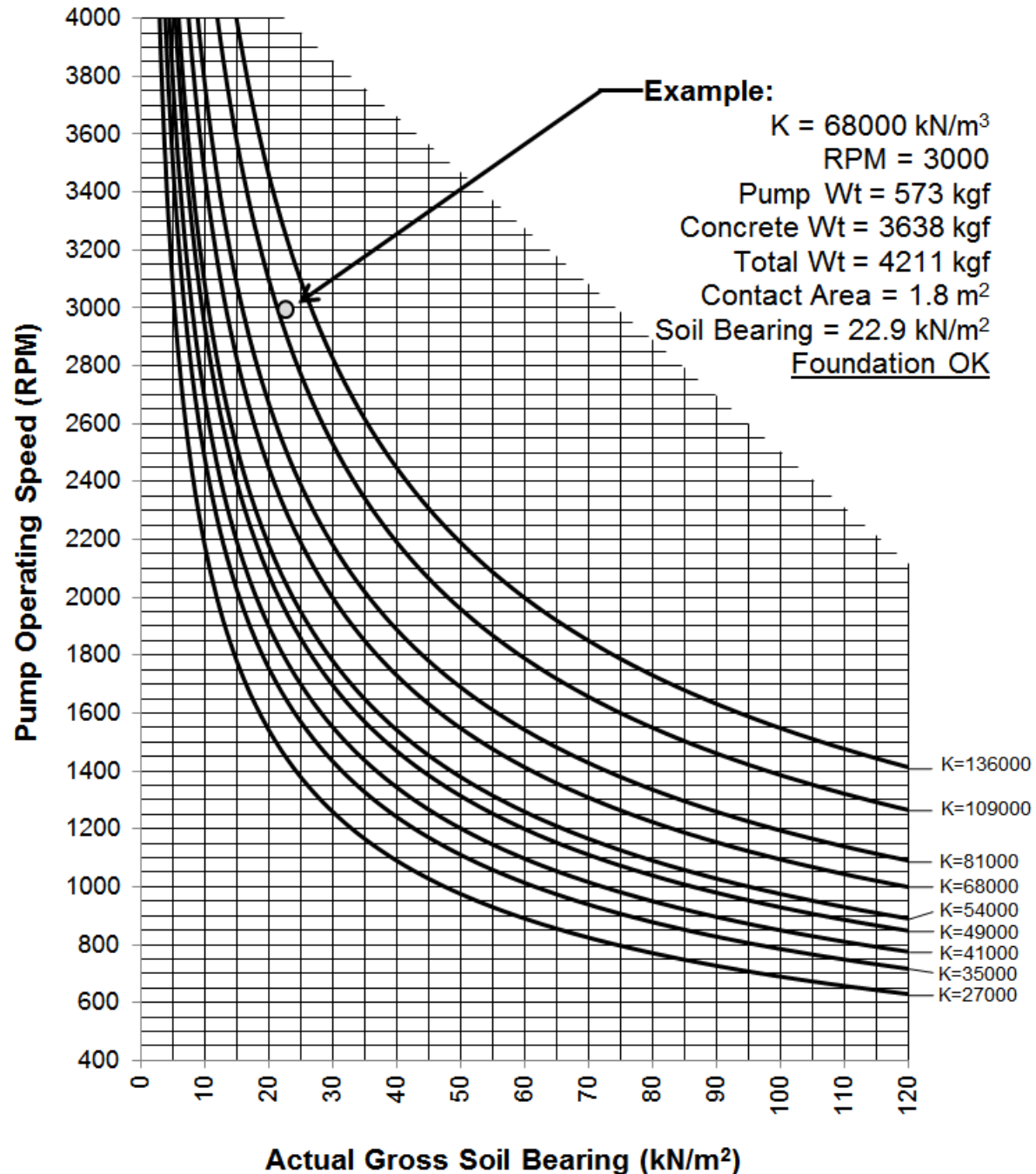




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Pump Foundation Design Curves [Imperial and Metric]

METRIC





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Sample Design 1: Pump Foundations [Imperial Units]

Design a foundation for a pump based information provided by Pump Vendor for a project site with well graded sand.

Pump Vendor Outline Information:

- Pump Base Size 84.5" x 30"
- Center to Center (c/c) Bolt Distance for Outermost Bolts 72.5" x 27"
- Anchor Bolt Holes in Pump Frame = 1", Assume Bolt Diameter 7/8"
- Number of Bolt Holes in Pump Frame = 8
- Pump Speed 3600 RPM

Design Data:

- Weight of Concrete 150 pcf

1. Select Modulus of Subgrade Reaction from the Attachment 1

$K = 250 \text{ lbs/in}^3$ for well graded sand

2. Compile Minimum Foundation Dimensions:

Using 7/8" diameter sleeved anchor bolts for 1" diameter bolt holes.

Bolt projection = 3/8" grip (skid flange) + 1" grout + 2 1/4" nuts = 3-5/8" = say 4"

Standard bolt length = 1'-6"

Minimum foundation size based on bolt c/c distance and edge distance

$$= (27" + 2 \times 6 \times 7/8") \times (72.5" + 2 \times 6 \times 7/8") = \underline{37.5"} \times 83"$$

Minimum foundation size based pump base size and oil pan protection

$$= (30" + 2 \times 2") \times (84.5" + 2 \times 2") = 34" \times \underline{88.5"}"$$

Depth of foundation = 9" above HPFS + 2'-0" below HPFS = 2'-9" (no frost requirement)

Therefore try foundation: 38" x 90" x 2'-9" deep (3'-2" x 7'-6" x 2'-9" deep)

3. Check Mass Ratio:

$$W_c \text{ (foundation)} = (3.17') \times (7.5') \times (2.75') \times (150 \text{ pcf}) = 9,807 \text{ lbs}$$

Total weight of pump, motor, and base $W_p = 2324 \text{ lbs}$

$$\text{Mass Ratio} = W_c / W_p = 9807 / 2324 = 4.2 > 3 \leftarrow \text{OK}$$

4. Check Frequency Ratio:

$F_{\text{pump}} = 3600 \text{ RPM}$ (motor speed from seller outline or data sheet)

$$F_n \text{ (foundation)} = 2253 \sqrt{\frac{K}{P_g}} \text{ where}$$

$$K = 250 \text{ lbs/in}^3$$



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Sample Design 1: Pump Foundations [Imperial Units]

$$P_g = (2324 + 9807) / (3.17 \times 7.5) = 510 \text{ lbs/ft}^2$$

$$F_n = 2253 \sqrt{\frac{250}{510}} = 2208 \text{ CPM}$$

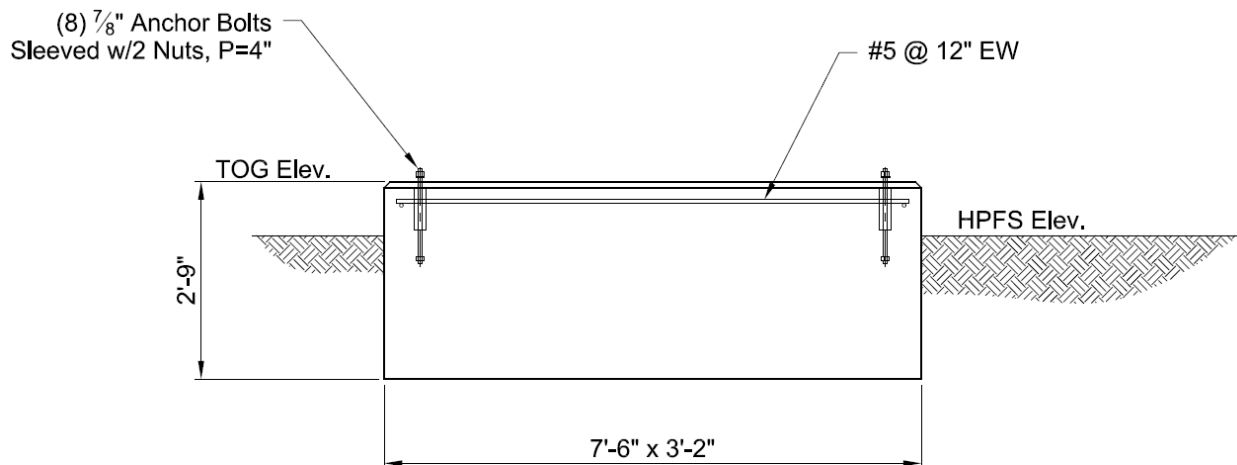
$$\text{Frequency Ratio } R = \frac{F_{\text{pump}}}{F_n} = \frac{3600}{1577} = 2.28 \geq 1.4 \quad \leftarrow \text{OK}$$

5. Reinforcing:

In lieu of any client requirements provide a minimum reinforcing of #5 at 12 c/c mat each way at top only.

6. Design Summary

Provide 3'-2" x 7'-6" x 2'-9" foundation with #5 at 12" each way at top with 8-7/8" Sleeved Bolts with 2 nuts and 4" projection



Pump Foundation Elevation



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Sample Design 1: Pump Foundations [Metric Units]

Design a foundation for a pump based information provided by Pump Vendor and for a project site with well graded sand.

Pump Vendor Outline Information:

- Pump Operating Speed 3000 RPM
- Pump Base Size 2150mm x 760mm
- Pump Total Operating Weight = 1100kgf
- Center to Center (c/c) Bolt Distance for Outermost Bolts 1800mm x 700mm
- Anchor Bolt Holes in Pump Frame = 25mm, Assume Bolt Diameter 22mm
- Number of Bolt Holes in Pump Frame = 8

Design Data:

- Weight of Concrete 2500kgf/m³

1. Select Modulus of Subgrade Reaction from the Attachment 1

$K = 68000 \text{ kN/m}^3$ for well graded sand

2. Compile Minimum Foundation Dimensions:

Using 22mm diameter sleeved anchor bolts for 25mm diameter bolt holes.

Bolt projection = 10mm grip (skid flange) + 25 mm grout + 60 mm nuts = say 100mm

Standard bolt length = 450mm

Minimum foundation size based on bolt c/c distance and minimum edge distance

$$= (700 + 2 \times 6 \times 22\text{mm}) \times (1800 + 2 \times 6 \times 22\text{mm}) = \underline{964\text{mm}} \times 2064 \text{ mm.}$$

Minimum foundation size based on pump base size and oil pan protection

$$= (760 + 2 \times 50\text{mm}) \times (2150 + 2 \times 50\text{mm}) = 860\text{mm} \times \underline{2250\text{mm}}$$

Depth of foundation = 250mm above HPFS + 600mm below HPFS = 850mm (no frost requirement)

Therefore try foundation: 1000mm x 2300mm x 900mm deep.

3. Check Mass Ratio:

$$W_c (\text{foundation}) = (1.0) \times (2.3) \times (0.9) \times (2500 \text{ kgf/m}^3) = 5175\text{kgf}$$

Total weight of pump, motor and base, $W_p = 1100\text{kgf}$

$$\text{Mass Ratio} = W_c / W_p = 5175 / 1100 = 4.7 \geq 3 \quad \leftarrow \text{OK}$$

4. Check Frequency Ratio:

$$F_{\text{pump}} = 3000 \text{ RPM (motor speed from seller outline or data sheet)}$$

$$F_n (\text{foundation}) = 29.9 \sqrt{\frac{K}{P_g}} \quad \text{where}$$

$$K = 68000 \text{ kN/m}^3$$



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Sample Design 1: Pump Foundations [Metric Units]

$$P_g = (5175 + 1100)(9.81) / (1.0 \times 2.3)(1000) = 27 \text{ kN/m}^2$$

$$F_n = 29.9 \sqrt{\frac{68000}{27}} = 1501 \text{ CPM}$$

$$\text{Frequency Ratio } R = \frac{F_{\text{pump}}}{F_n} = \frac{3600}{1501} = 2.40 \geq 1.4 \quad \leftarrow \text{OK}$$

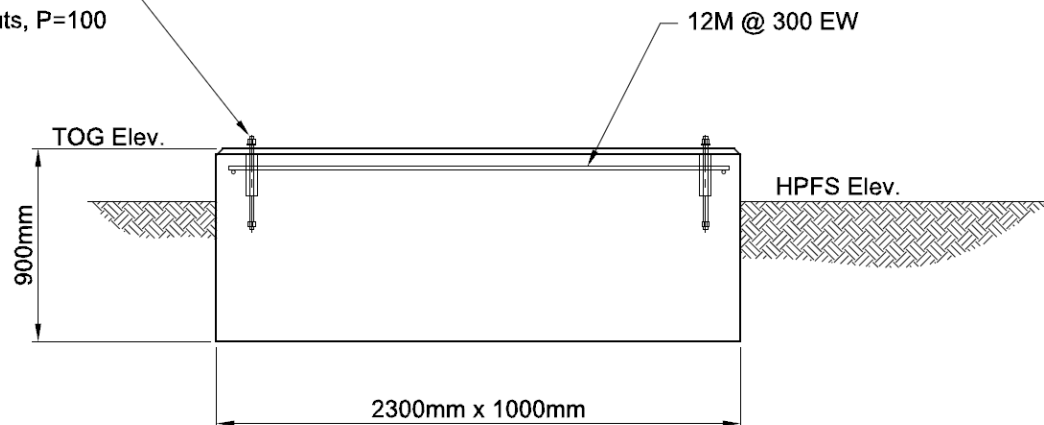
5. Reinforcing:

In lieu of any client requirements provide a minimum reinforcing of 12mm at 300mm c/c mat each way at top only.

6. Design Summary:

Therefore provide 1000mm x 2300mm x 900mm foundation with 12M at 300mm each way at top with 8-22mm Sleeved Bolts with 2 nuts and 100mm projection

(8) M22 Anchor Bolts
Sleeved w/2 Nuts, P=100

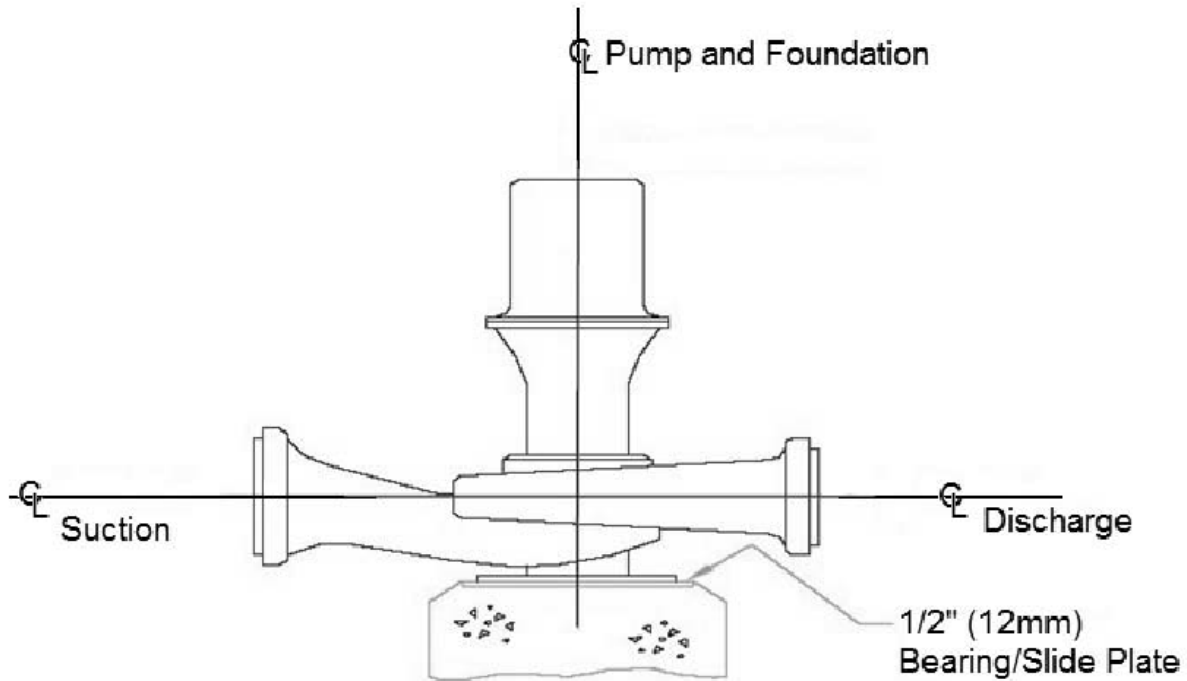


Pump Foundation Elevation



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In-line Pump Foundation Design Criteria



In-Line Pump Foundation

Notes:

1. In-Line Pumps are to be supported as required by pump manufacturer and/or by pipe stress engineer.
2. For pumps requiring vertical support only no fastening between pump and slide plate or between concrete and slide plate is required.
3. Coordinate lines for all in-line pumps shall be centerline of pump, not centerline of discharge
4. Edge of foundation shall extend a minimum of 2\" (50mm) and a maximum of 3\" (75mm) beyond the pump base.
5. Bearing/slide plate shall extend 1\" (25mm) minimum beyond the pump base.
6. Overall dimensions of the foundation (width and length) should be rounded up to the nearest even inch (For metric design, follow direction of discipline lead engineer, as round up values vary from project to project)